



Enhanced Autonomic Fear Conditioning at Age 3 in Successful Adult Criminal Offenders

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Abstract

There is a dearth of data on “successful” criminal offenders who escape detection, and no research on emotion processes characterizing this group. This study examines whether autonomic fear conditioning at age 3 years is associated with adult successful and unsuccessful (caught) criminal offending. Participants (N=940) were drawn from a birth cohort on the island of Mauritius. Autonomic fear conditioning was assessed at age 3, with skin conductance recorded to CS+ (reinforced) and CS- (unreinforced) auditory stimuli. Self-report criminal offending was assessed at age 39. Offenders were divided into successful and unsuccessful groups based on detection/conviction at ages 23 and 39, and compared to non-criminal controls. There was a significant group x conditional stimulus interaction ($p = .019$). A breakdown of this interaction indicated that successful criminals showed significantly greater responding to the CS+ than both unsuccessful criminals ($p=.003$) and to controls ($p<.001$), with no group differences on CS-, indicating superior conditioning in successful criminals. Three sensitivity analyses confirmed these findings. We hypothesize that enhanced fear conditioning in successful offenders results in an enhanced ability to detect environmental cues associated with punishment, resulting in escape from law enforcement agencies.

Keywords

successful criminals; fear conditioning; skin conductance; psychopathy

1. Introduction

Criminal offending is a key characteristic of antisocial personality disorder, while other clinical disorders have raised rates of offending, including borderline personality disorder (Sansone & Sansone, 2009), ADHD (Young & Cocallis, 2021), and schizophrenia-spectrum disorders (Whiting et al., 2022). As such, an understanding of individual differences which predispose to criminal offending is of significance to researchers and practitioners alike.

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One influential emotion theory of antisocial behavior and offending consists of poor fear conditioning (Birbaumer et al., 2005; Chen, 2020; Gao et al., 2010a). Socialized individuals are hypothesized to learn to avoid antisocial and criminal acts by successfully associating otherwise neutral stimuli with later socializing punishments (Eysenck, 1977). This association learning is hypothesized to result in an increase in anticipatory fear whenever the individual contemplates the commission of an antisocial act, which in turn provides motivation to avoid such stimuli and thus the commission of antisocial, rule-breaking behavior (Blair, 2007; Patrick, 2017). Poor fear conditioning in turn is hypothesized to be associated with a lack of anticipatory fear and consequently increased antisocial behavior. In support of this theory, multiple studies have documented poor fear conditioning and deficient fear in antisocial populations (Birbaumer et al., 2005; Chen et al., 2021; Gao et al., 2010a; Patrick et al., 2012; Raine et al., 1996).

Within the community there is significant heterogeneity in offending. In particular, while apprehended offenders have been quite extensively researched, much less is known about offenders who are successful in evading detection. Indeed, to the authors' knowledge, there has been no published empirical research on successful crime – those who commit crime but are not caught. There has however been one recent psychosocial study on successful delinquents. Successful delinquents - defined as those who self-reported offended between ages 10 and 18 but who were not convicted at age 26 - were characterized by low risk-taking, intact families, and having non-criminal parents (Farrington et al., 2023). While this study provides important insights into psychosocial explanations of successful delinquency, research on adult crime is lacking, and there has been no examination in any antisocial grouping of fear conditioning, or indeed any other biological variable.

The current study aims to address this gap in the literature within the context of a longitudinal birth cohort study that we have been conducting for the past 53 years (Raine et al., 2010). Following the design of the successful delinquent study (Farrington et al., 2023), we classified adult criminal offenders into “successful” (uncaught) and “unsuccessful” (caught) criminal groups based on both self-report and court record data collected up to age 39. Based on the prior literature, we hypothesized that unsuccessful offenders would be characterized by poor fear conditioning. In contrast, we hypothesized that successful offenders who evade detection may be more sensitive to cues that predict detection and punishment, and in comparison to caught offenders would show better fear conditioning.

2. Methods

2.1 Participants.

A total of 940 participants from a larger longitudinal birth cohort on child health and development were included in the current study. Participants in this community sample consisted of 535 male (54.6%) and 441 female (45.0%) adults aged 39 years (mean = 38.99, SD = 1.33) from the island of Mauritius, an African tropical island in the Indian Ocean. Further details on this birth cohort study are available in Raine et al. (2010). Ethnicity was as follows: 72.6% Indian, 27.4% Creole. Written informed consent was obtained from participants. Those with missing data differed to those with complete data on gender and social adversity (see Supplement) and so these variables were controlled for in all analyses.

IRB approval was obtained at early data collection stages from the University of Southern California and at later stages from the University of Pennsylvania.

2.2 Fear Conditioning at Age 3 Years

Recording. Electrodermal data were collected using a Grass Type 79 polygraph with a constant voltage system, Beckman miniature silver/silver chloride electrodes, and an electrolyte consisting of 0.5% KCl in 2% agar-agar. Electrodermal activity was recorded from the medial phalanges of the index and middle fingers of the left hand (Venables, 1978).

Conditioning paradigm. The conditioning paradigm consisted of 12 tones (9 CS+ and 3 CS-) with a 66% partial-reinforcement schedule (see Supplemental Figure S1). The CS+ was a 1000 Hz, 60 dB, 12.5 s tone with rise and fall time of 25 ms, while CS- was a 500 Hz, 60 dB, 12.5 s tone with rise and fall time of 25 ms. The UCS was an auditory stimulus of 90 dB intensity and 4.5 s duration recorded from white noise played in a tin can with metal jangling objects to add unpleasant low and high frequency components to the stimulus. Stimuli were presented in a fixed order. The inter-stimulus interval was 10 s (onset to onset), with a randomized inter-trial interval of 38 s (range 34–42 s). Consistent with our prior work (Gao et al., 2010a, 2010b, 2010c), the three CS- trials and three CS+ trials which were contiguous in time were utilized. Electrodermal responses greater than 0.05 μ S within a latency window of 1–3 s after the onset of the CS were scored. Trials were averaged within a class (CS+, CS-) to enhance reliability and to form overall measures of responding to reinforced (CS+) and unreinforced trials (CS-) respectively (see Gao et al. 2010a for further details). The paradigm was kept short as from a human subjects perspective we had to limit the time three-year-olds were exposed to aversive stimuli, and children were sat on the mother's lap throughout the paradigm for comfort.

2.3 Successful and Unsuccessful Criminals.

The participants at age 39 (N=940) were divided into three groups: Non-Criminal Controls (N=500), Successful Criminals (N=238) and Unsuccessful Criminals (N=202). Groups were formed based on a self-report crime interview at age 39, together with court conviction data collected at age 23. Non-Criminal Controls reported no crime at age 39 and had no convictions at age 23. Unsuccessful criminals reported perpetrating crime at age 39 and also that they had been questioned by the police, or were arrested or convicted for their crimes, or else had a court conviction at age 23. Successful criminals reported having perpetrated crimes at age 39 but had not been detected for them and had also not been convicted at age 23.

The self-report crime methodology has been extensively validated (Gomes et al., 2019; Jolliffe et al., 2003) and provides a more encompassing measure of crime than official convictions, with research showing that for every official conviction, an average of 39 crimes were perpetrated based on self-reports (Farrington et al., 2023). At age 23 years, official court records were searched for registration of offenses that included property, drug, violence, and serious driving offenses.

2.4 Covariates

A number of potential confounds were controlled for in analyses reported below. Gender and ethnicity were collected in a demographic interview at age 39. Social adversity data at age 3 was collected by social workers who visited the homes of the children at age 3 years (Raine et al., 2002). A total adversity score was created by adding one point for each of the following nine variables: father uneducated (no schooling, 30.0%), mother uneducated (no schooling, 29.4%), semi-skilled or unskilled occupation (occupational status 3 or less on an 8-point occupational scale: 0 = unemployed, 4 = factory worker, cook, 8 = academic, head of large business, 55.5%), teenage mother (age 19 or younger when child was born, 14.2%), single parent status (2.1%), separation from parents (orphaned or raised by substitute mother, 0.9%), large family size (sibling order 5th or higher by age 3 years, 30.0%), poor health of mother (coded 1 on a 3-point scale: above average = 3, average = 2 below average = 1, 3.3%), overcrowded home (5 or more family members / house room, 28.8%). Scores ranged from 0 to 7 with a mean of 1.94 and SD of 1.39. Autonomic control variables consisted of skin conductance level and number of spontaneous (i.e. non-stimulus driven) skin conductance responses recorded during the rest period. Antisocial controls consisted of both number of self-report crimes and psychopathic personality assessed using the Psychopathic Personality Inventory (Lilienfeld & Andrews, 1996).

Transparency and Openness.

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study, and the study follows JARS (Appelbaum, et al., 2018). All data, analysis code, and research materials are available at <https://doi.org/10.7910/DVN/OQPSTI> for the purpose of checking findings. Data were analyzed using SSPP version 29.0.1.1 This study's design and its analysis were not pre-registered.

2.5 Statistical analyses.

A complete list of main study variables is given in Table 1. A repeated measures analysis of covariance with stimulus type (CS+, CS-) as the within-subject factor and criminal grouping as the between-subjects factor was conducted. Paired sample t-tests were conducted to break down the interaction. All tests are two-tailed except where otherwise stated. Sample size was based on all participants who responded to request for follow-up at age 39. There were no data exclusions.

3. Results

3.1 Main analysis.

Controlling for demographic confounds (gender, ethnicity, and early social adversity), there was a main effect of group, $F(1,934) = 4.32, p = .014$, and a significant group x stimulus interaction, $F(2,934) = 4.00, p = .019$. Results are illustrated in Figure 1 where it can be seen that while both successful criminals and controls showed higher values of CS+ compared to CS- (suggesting conditioning), this was not shown in unsuccessful offenders.

Breaking down this interaction, paired samples t-tests comparing CS+ with CS- amplitudes on each individual group indicated that unsuccessful criminals failed to show fear

conditioning ($t = -0.27$, $p = .79$), controls trended towards showing fear conditioning ($t = 1.57$, $p = .058$, one-tailed), while successful criminals showed significant fear conditioning ($t = 3.23$, $p = .001$).

3.2 Sensitivity analyses.

We examined the robustness of findings using three sensitivity analyses as follows:

Independent samples t-tests. In contrast to the paired samples t-tests outlined above, we examined differences between groups on each of the conditioned stimuli. Groups did not differ significantly on their responses to the CS- (all $p > .18$). In contrast, successful criminals showed significantly higher values to the CS+ compared to both unsuccessful criminals ($p = .003$) and also controls ($p < .001$).

Controlling for autonomic confounds. To evaluate whether differences in fear conditioning were a function of autonomic arousal, skin conductance levels and spontaneous skin conductance responses during rest were entered as covariates. The group x stimulus interaction remained significant, $F(2,914) = 4.04$, $p = .018$. Furthermore, after adding to these covariates control for response amplitude to the UCS, the group x stimulus interaction remained significant, $F(2,913) = 3.91$, $p = .02$.

Controlling for antisocial confounds. While both offender groups committed crime, the unsuccessful group committed more crimes than the successful group ($t = 4.83$, $p < .001$) and had higher psychopathic personality scores ($t = 2.70$, $p = .007$). After controlling for these group differences, the group x stimulus interaction remained significant, $F(2,834) = 4.37$, $p = .013$.

Trial analysis. While we had averaged trials across stimulus type to provide a more reliable CS+/CS- measures, we also analyzed data on a trial-by-trial basis in a repeated measures MANCOVA with trials and stimuli as repeated measures, again controlling for covariates. There was a main effect of stimulus, $F(1,934) = 9.2$, $p = .003$, indicating larger responses to the CS+ than the CS-. There was also a group x stimulus interaction, $F(2,934) = 3.73$, $p = .024$, again indicating significantly larger responses to the CS+ compared to the CS- in the successful criminals, but only marginal effects in controls and null effects in Unsuccessful Criminals. No other main effects or interactions were significant.

4. Discussion

We set out to examine whether successful and unsuccessful criminals differ on fear conditioning. We found that in contrast to unsuccessful criminals who failed to show fear conditioning at age 3, successful criminals evidenced superior fear conditioning. This relationship spanning 36 years could not be explained by demographic, autonomic, or antisocial confounds. Findings are the first to examine successful criminal offending and suggest that better fear conditioning in this sub-group of offenders results in an enhanced ability to detect environmental cues associated with punishment, resulting in escape from law enforcement agencies.

The finding that unsuccessful, caught offenders show poor fear conditioning is consistent with a relative strong body of evidence documenting poor fear conditioning in antisocial populations (Birbaumer et al., 2005; Chen et al., 2021; Gao et al., 2010a; Patrick et al., 2012; Raine et al., 1996). The more striking finding of this study, however, is the significant fear conditioning in the successful, uncaught offenders. Indeed, this group showed superior conditioning not just to the unsuccessful criminal offenders, but also to the controls. Fear conditioning reflects the ability to form an association between a neutral environmental cue and an aversive event. When engaging in crime, successful offenders may be more sensitive to benign cues in their environment that provide a forewarning of detection/threat or the possible presence of law enforcement agents, evoking anticipatory fear which motivates them to evade capture (Chen, 2020). With this sensitivity diminished in unsuccessful offenders, they would be hypothesized to lack this anticipatory fear and consequently have an increased probability of detection.

Fear conditioning in this sample took place in the presence of the mother. There is a significant animal literature (Callaghan et al., 2019) as well as a growing human literature (Gunnar et al. 2015) documenting that parental presence during fear conditioning provides a buffering of fear-related behaviors. For example, four-year-old children who electrodermally condition in the presence of their mother, compared to those undergoing conditioning alone, show a behavioral approach bias *in favor* of the CS+ compared to the CS-, even though the CS+ is rated as being disliked (Tottenham et al., 2019). To the extent that the better conditioning in successful criminals reflects an impairment in early maternal buffering, such individuals would be more likely to behaviorally avoid contexts and situations (CS+) that have been paired with punishment, again potentially reducing the likelihood of detection from law enforcement agencies.

Fear conditioning is predicated on a number of brain regions, including the amygdala, insula, and anterior cingulate (Sehlmeyer et al., 2009), and more broadly a central autonomic-interoceptive network of relevance to brain-body interactions and subjective emotional awareness (Fullana et al., 2016). The current findings provide some indirect but initial clues as to which brain regions may be implicated in successful and unsuccessful offending. Meta-analysis of fMRI data indicates that processing of the “safety signal” (the CS-) involves somatosensory cortex, ventromedial prefrontal cortex, posterior cingulate, and lateral orbitofrontal cortex (Fullana et al., 2016). These areas appear not to be relevant given that groups did not differ on responding to the CS-. Instead, group differences emerged specifically in processing the CS+, processing that has implicated the anterior insula, the ventral striatum, pre- and post-supplementary motor areas, the left dorsal anterior cingulate, dorsal prefrontal cortex, and lateral premotor cortex (Fullana et al., 2016). Whether or not future neuroimaging studies show that successful criminals (who showed significantly greater responding to the CS+ than both unsuccessful criminals and controls) have enhanced functioning in these same regions remains to be seen.

Sensitivity analyses largely confirmed findings with the exception of the trial-by-trial analysis. This documented conditioning through a main effect for stimulus (CS+ > CS-) as well as a significant group x stimulus interaction. However, no stimulus x trial interaction or stimulus x trial x group interaction was observed. Prior electrodermal conditioning studies

of children predominantly either fail to observe these interactions (Tottenham et al., 2019) or do not report these interactions (Reinhard et al., 2022; Schiele et al., 2016; McLaughlin et al., 2016; Fairchild et al., 2008), and instead rely on CS+/CS- responses averaged over trials to support a claim of conditioning. While conditioned skin conductance responses are sensitive to CS-US contingencies, they have characteristics of an orienting response that habituate (Ohman, 2000), unlike the gradual response acquisition that is typically expected of a conditioned response (Ohman, 1983; Ohman, 2000). While averaging conditioned responses over trials provides a more reliable index than single-trial electrodermal data, future studies using more conditioning trials and more robust measures of fear conditioning are better placed to confirm or question the robustness of the current findings.

Limitations need to be acknowledged. Measuring autonomic fear conditioning in a large sample of three-year-old children is challenging, and ethical constraints precluded the use of an electric shock as a UCS. The fact that the UCS was only mildly aversive, coupled with the partial reinforcement paradigm, resulted in the control group showing only a trend toward conditioning, although for the entire birth cohort sample, conditioning did take place ($t = 3.62$, $p < .001$). Findings from this non-Western sample require replication in Western samples. Representativeness of this sample to the entire country of Mauritius cannot be claimed. We also caution that causal inferences cannot be drawn from these data, although the longitudinal nature of these findings starting in early life helps to tease out the temporal ordering of the constructs concerned.

In conclusion, this study highlights a new paradigm – successful and unsuccessful criminality – to more comprehensively understand the emotion etiology of antisocial behaviors that are a serious comorbid condition of a number of clinical disorders, and document early fear conditioning as one neurobiological variable of interest. Because some studies have shown that autonomic functioning moderates treatment outcome in antisocial individuals, with reduced autonomic functioning showing less improvement and those with higher autonomic functioning showing better improvement (Cornet et al., 2014), the current findings have potential treatment and risk prediction implications. Knowledge of both risk factors and interventions for this uncharted condition can be of significance more broadly given the comorbidity of crime with mental illness, and the likelihood that individuals presenting with clinical disorders may have significant rates of successful crime. From a basic affective science perspective, future developmental brain imaging research employing a fear conditioning paradigm could help better elucidate the brain mechanisms that subserve this differentiating factor in successful and unsuccessful criminals.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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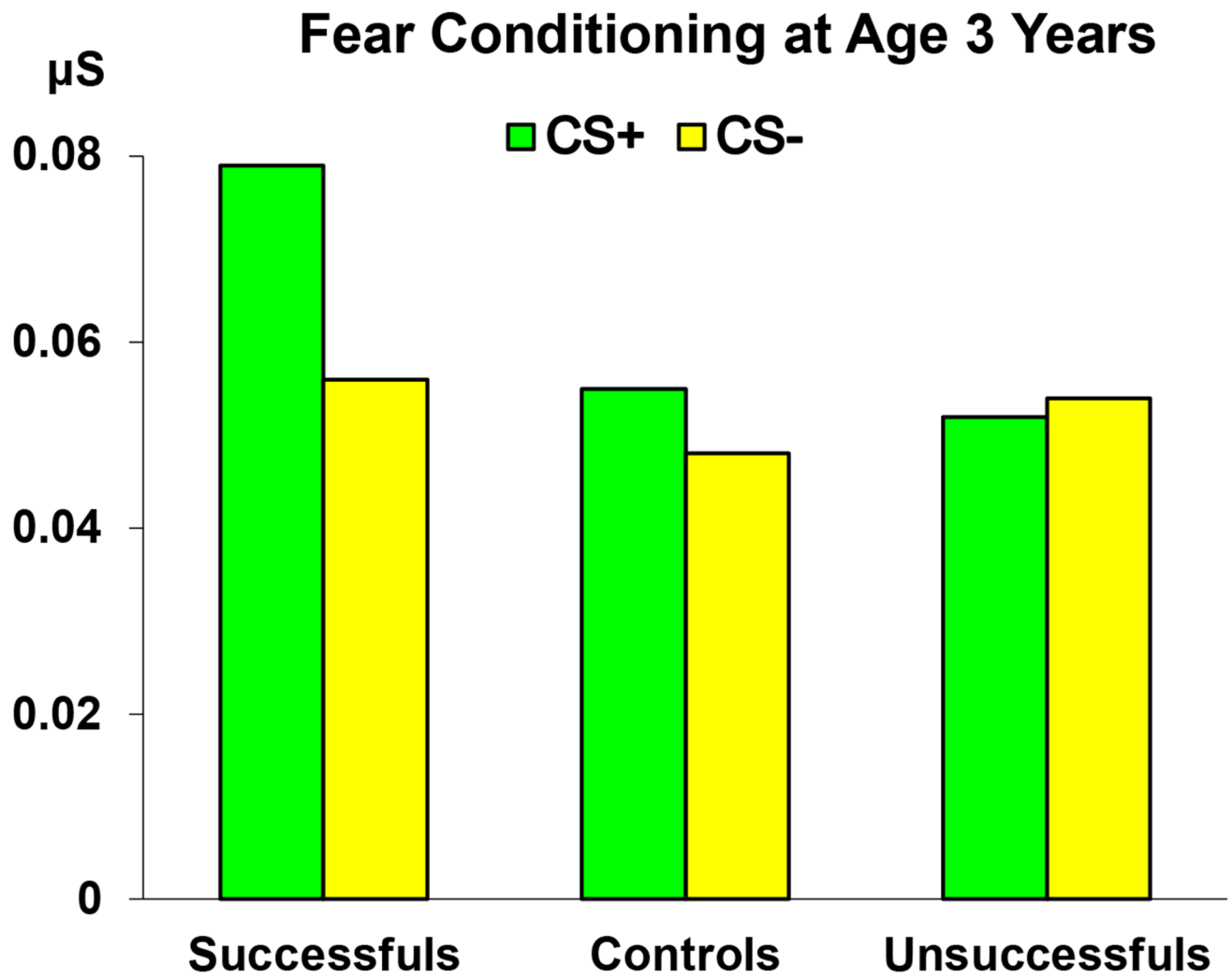


Figure 1.

Skin conductance responding at age 3 years to the CS+ (reinforced) and CS- (unreinforced) stimuli in the three groups, illustrating superior conditioning in successful criminals. Greater responding to the CS+ relative to the CS- is an indicator of conditioning.

Table 1.

Characteristics of successful criminals, unsuccessful criminals, and noncriminal controls on main study variables.

	Successful Criminals (N = 238)		Unsuccessful Criminals (N = 202)		Controls (N = 500)	
	N	%	N	%	N	%
Male	171	71.8%	192	95.1%	172	34.4%
Creole	78	32.8%	67	33.2%	113	22.6%
	Mean	SD	Mean	SD	Mean	SD
Age	38.61	0.62	38.87	0.95	39.21	1.60
Social Adversity	1.92	1.34	2.05	1.48	1.95	1.29
Self-report crime	4.33	4.54	7.42	9.46	0	0
Psychopathy	121.07	12.90	124.33	12.23	114.9	12.87
CS+	0.08	0.12	0.05	0.09	0.06	0.09
CS-	0.06	0.09	0.05	0.08	0.05	0.08
Unconditional Stimulus	.19	.25	.18	.19	.19	.22
Spontaneous Responses	2.14	2.38	1.73	2.00	2.34	2.59
Skin Conductance Levels	2.39	1.42	2.53	1.75	2.46	1.61